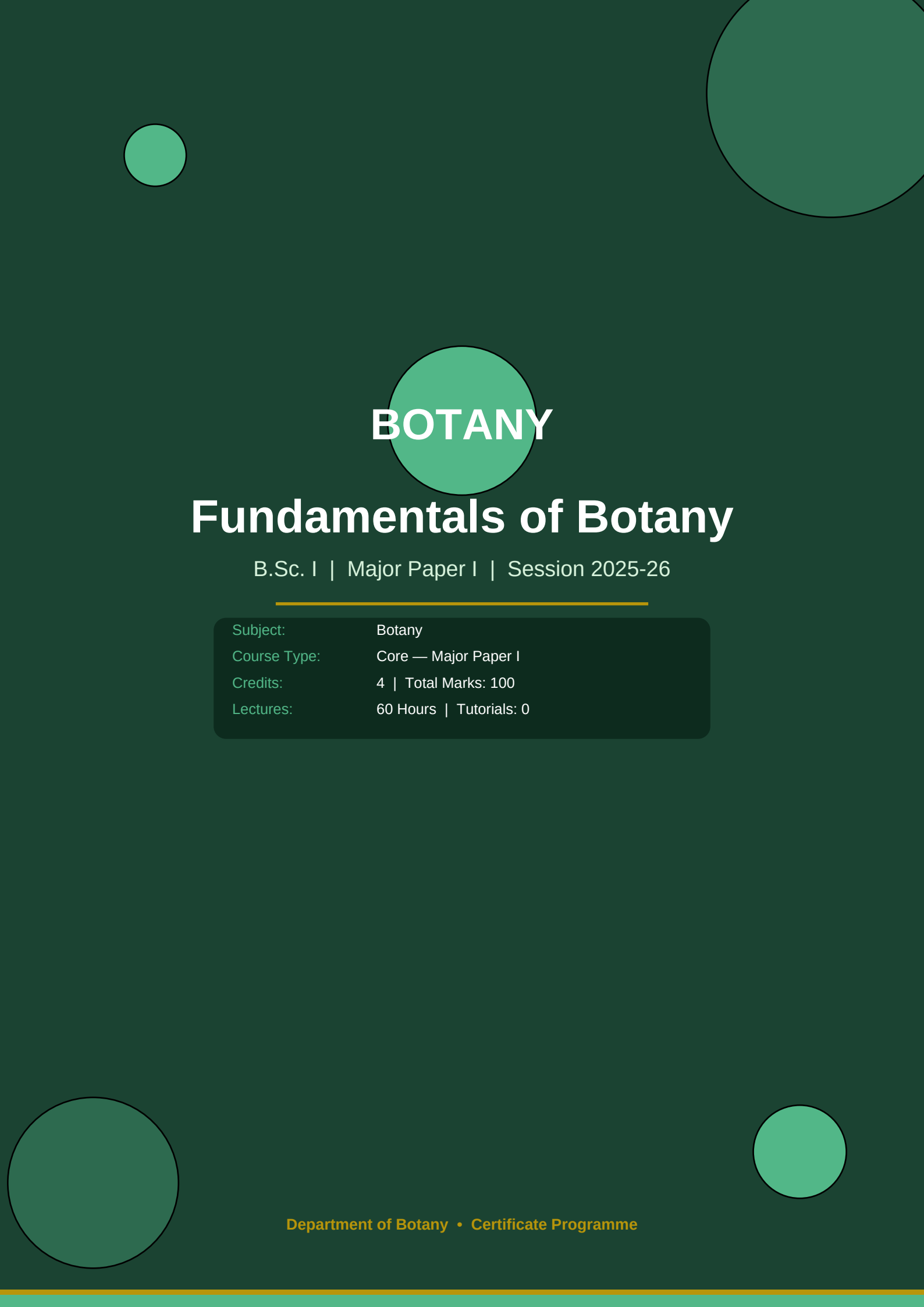




BOTANY

Fundamentals of Botany

B.Sc. I | Major Paper I | Session 2025-26

Subject:	Botany
Course Type:	Core — Major Paper I
Credits:	4 Total Marks: 100
Lectures:	60 Hours Tutorials: 0

UNIT I | Plants in "Bhartiya Gyan Parampara" (12 Lectures)

1.1 Botany — Definition, Scope and Importance

Definition: Botany (Greek *Botane* = plant) is the branch of Biology dealing with the scientific study of plants — their structure, growth, reproduction, metabolism, diseases and evolutionary relationships. Also called **Plant Biology** or **Phytology**.

Scope — Major Branches:

Branch	What it Studies
Morphology	External form and structure of plants
Anatomy	Internal structure of plant organs
Physiology	Life functions and metabolic processes
Taxonomy	Classification and naming of plants
Ecology	Plants and their environment
Genetics	Heredity and variation in plants
Biochemistry	Chemical processes within plants
Ethnobotany	Plants in human culture and tradition
Economic Botany	Plants of commercial/industrial value

Importance:

- **Food Source:** All food chains begin with plants (rice, wheat, pulses, vegetables).
- **Oxygen:** Photosynthesis by plants maintains atmospheric O₂ balance.
- **Medicine:** >80 % of medicines are plant-derived (Quinine, Morphine, Aspirin).
- **Economy:** Timber, rubber, cotton, jute, spices, perfumes — all from plants.
- **Environment:** Plants prevent soil erosion, regulate rainfall and absorb CO₂.
- **Research:** Mendel used pea plants to discover genetics; biofuels from plants.

1.2 History of Botany in India

Period	Key Developments
Ancient (Vedic) 1500-500 BCE	Plants described in Rigveda & Atharvaveda; classification into trees, shrubs, herbs; Parashar Vrikshayurveda; Charaka & Sushruta Samhita
Medieval (Mughal Era)	Plant gardens (Bagicha) developed; Ain-i-Akbari records plant uses; interest in horticulture
British Period 18th–19th c.	William Roxburgh — "Father of Indian Botany", wrote Flora Indica; J.D. Hooker — Flora of British India (17,000+ spp.)
Modern (Post-1947)	Botanical Survey of India (Kolkata); CSIR/ICAR research; Green Revolution HYV crops; J.C. Bose proved plant sensitivity

1.3 Botanical Concepts in Vedas, Upanishads, Ramayan and Puranas

- Rigveda (~1500 BCE)**
Classifies plants as *Vanaspati* (trees with fruit, no visible flower), *Vriksha* (trees with flower + fruit), *Oshadhi* (medicinal herb) and *Virudh* (climber). The sacred Soma plant is prominently described.
- Atharvaveda**
First systematic text on medicinal plants; 290+ hymns on plants; describes Apamarga, Durva, Pippali; introduces the concept of *Bheshaj* (medicine).
- Upanishads**
Chandogya Upanishad discusses plant consciousness; *Brihadaranyaka Upanishad* states plants are living beings with a soul (Atma).
- Ramayan (Valmiki)**
Mentions 100+ plant species — Ashoka Vatika (*Saraca asoca*), Panchavati forest (Banyan, Peepal, Bel, Amla, Ashoka), Sanjeevani Booti, Jamun, Mango, Coconut.
- Puranas**
Vishnu Purana lists five plant types: Vriksha, Gulma, Lata, Valli, Truna. *Agni* and *Garuda Purana* describe medicinal plants; *Brahma Purana* lists sacred Panchvati trees.

1.4 Contribution of Indian Rishis

Rishi	Key Work	Major Contributions
Parashar	Vrikshayurveda	Oldest plant-science text; classified Dicots (Dwimatruka) & Monocots (Ekamatruka); described plant diseases & treatments
Charak	Charaka Samhita	50 Mahakashaya groups; concept of Dravyaguna (plant properties); different plant parts have different medicinal values
Sushrut	Sushruta Samhita	Classified Vanaspati/Vriksha/Virudh/Oshadhi; described 700+ medicinal plants; used plant fibres as surgical sutures
Surpaal	Vrikshayurveda (Horticulture)	Propagation techniques; treatment of plant diseases with herbal preparations; focus on crop improvement
Kautilya	Arthashastra	Plants of economic & strategic importance; forest conservation as state policy; plant-based medicines & antidotes

1.5 Plants in Traditional Food and Folk Medicine

Spices with Dual Role (Food + Medicine):

Plant	Scientific Name	Medicinal Use
Turmeric	Curcuma longa	Anti-inflammatory, antiseptic, wound healing
Ginger	Zingiber officinale	Digestive aid, nausea, cold & cough
Garlic	Allium sativum	Antibiotic, lowers cholesterol, heart health
Neem	Azadirachta indica	Skin diseases, diabetes, malaria, dental care

Tulsi	Ocimum tenuiflorum	Cough, cold, fever, anti-stress adaptogen
Giloy	Tinospora cordifolia	Immunity booster, dengue fever, diabetes
Amla	Phyllanthus emblica	Richest Vit. C source, hair & eye health
Ashwagandha	Withania somnifera	Stress reducer, improves stamina, insomnia



UNIT II | Morphology — External Structure of Plants (12 Lectures)

1.1 Basic Concepts — Thallus vs Plant Body

Feature	Thallus	Plant Body (Cormus)
Differentiation	Absent — body undivided	Root, Stem, Leaf distinct
Vascular tissue	Absent	Present (xylem + phloem)
Examples	Spirogyra, Ulva, Marchantia	Ferns, Gymnosperms, Angiosperms
Level	Lower plants (algae, fungi)	Higher plants

1.2 Modifications of Roots

Root is the **underground, non-green, positively geotropic** part developing from the radicle. It anchors the plant and absorbs water + minerals. Has no nodes, internodes or leaves. Root tip is protected by the **root cap**; absorption occurs through **root hairs**.

Modification	Type	Example
Fusiform	Food storage (tap root)	Radish (<i>Raphanus sativus</i>)
Napiform	Food storage (tap root)	Turnip (<i>Brassica rapa</i>)
Conical	Food storage (tap root)	Carrot (<i>Daucus carota</i>)
Tuberous	Food storage (adventitious)	Sweet potato
Prop roots	Mechanical support	Banyan (<i>Ficus benghalensis</i>)
Stilt roots	Mechanical support	Maize, Sugarcane
Pneumatophores	Respiration (breathing)	Rhizophora, Avicennia (mangroves)
Nodulated roots	N ₂ fixation	Pea, Bean, Groundnut (<i>Rhizobium</i>)
Haustorial roots	Parasitism	Cuscuta (Amarbel)
Epiphytic roots	Moisture absorption from air	Vanda (orchid)

1.3 Modifications of Stem

Type	Category	Description	Example
Rhizome	Underground	Horizontal; nodes & internodes present	Ginger, Turmeric, Lotus
Tuber	Underground	Swollen tip; "eyes" = buds	Potato
Bulb	Underground	Condensed stem; fleshy scale leaves	Onion, Garlic
Corm	Underground	Solid, vertical, swollen stem	Colocasia, Gladiolus

Runner	Sub-aerial	Creeps on soil; roots at nodes	Grass, Strawberry
Stolon	Sub-aerial	Arches and touches ground	Jasmine, Mint
Tendrils	Aerial	Slender; helps climb	Cucumber, Passion flower
Thorn	Aerial	Hard, sharp; protection	Citrus, Bougainvillea
Phylloclade	Aerial	Flattened green stem; photosynthesis	Cactus (Opuntia)

1.4 Modifications of Leaf

Modification	Purpose	Example
Spines	Protection from herbivores	Cactus (leaves → spines), Barberry
Leaf tendrils	Climbing support	Pea (<i>Pisum sativum</i>) — leaflets become tendrils
Fleshy leaves	Water / food storage	Onion scale leaves, Aloe vera
Pitcher (<i>Nepenthes</i>)	Insect trapping (N-poor soil)	Pitcher plant
Phyllode	Photosynthesis (petiole flattened)	Australian <i>Acacia</i> spp.
Foliar buds	Vegetative reproduction	<i>Bryophyllum</i> — plantlets at leaf margins

1.5 Flower, Inflorescence, Fruits and Seeds

A **flower** is the reproductive structure of angiosperms consisting of four whorls: Calyx (sepals), Corolla (petals), Androecium (stamens — male) and Gynoecium (pistil — female).

Fruit Type	Characteristics	Examples
Drupe	Fleshy; hard stone (endocarp)	Mango, Coconut, Plum
Berry	Fleshy; no hard inner layer	Tomato, Grape, Banana
Legume/Pod	Dry; dehiscent along two sutures	Pea, Bean, Mustard
Capsule	Dry; many seeds; various dehiscence	Cotton, Poppy, <i>Datura</i>
Aggregate	Many free carpels of one flower	Raspberry, Strawberry
Multiple/Composite	Entire inflorescence	Jackfruit, Pineapple, Mulberry

UNIT III | Anatomy — Internal Structure of Plants (12 Lectures)

1.1 Plant Cell — Structure

Cell Theory: Schleiden (1838) + Schwann (1839) + Virchow (1855) | "Omnis cellula e cellula"

Organelle	Unique to Plants?	Key Function
Cell Wall	YES	Protection, shape, support; cellulose (primary) + lignin (secondary)
Chloroplast	YES	Photosynthesis; contains chlorophyll; own DNA
Large Vacuole	YES (large)	Turgor pressure, storage of pigments/waste/water
Plasma Membrane	No	Selective permeability; fluid mosaic model (Singer & Nicolson 1972)
Nucleus	No	Control centre; contains DNA; double membrane with pores
Mitochondria	No	ATP production; cellular respiration; own DNA
Ribosome	No	Protein synthesis (80S cytoplasm; 70S in plastids)
Golgi Apparatus	No	Packaging & secretion; makes cell wall pectin/hemicellulose
ER (Rough/Smooth)	No	Rough: protein synthesis; Smooth: lipid synthesis

1.2 Tissues — Classification

Tissue	Sub-type	Features	Function / Example
MERISTEMATIC	Apical	Tip of root/shoot; small cells	Primary growth (length)
MERISTEMATIC	Lateral	Vascular cambium, cork cambium	Secondary growth (girth)
MERISTEMATIC	Intercalary	Base of internode	Growth of grass internodes
Simple Permanent	Parenchyma	Living; thin walls; large vacuoles	Storage, photosynthesis, gas exchange
Simple Permanent	Collenchyma	Living; corner thickenings (cellulose)	Flexible support — celery stalk
Simple Permanent	Sclerenchyma	Dead; lignified; fibres & sclereids	Rigid support — jute, walnut shell
Complex Permanent	Xylem	Tracheids, vessels, fibres, parenchyma	Water & mineral conduction upward
Complex Permanent	Phloem	Sieve tubes, companion cells, fibres	Food conduction (source to sink)

1.3 – 1.5 Anatomy Comparison: Root, Stem, Leaf

Feature	Dicot Root	Monocot Root	Dicot Stem	Monocot Stem
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Epidermis	Epiblema with root hairs	Same; well-developed	Cuticle + stomata	Thick cuticle
Hypodermis	Absent	Absent	Collenchyma	Sclerenchyma
Vascular bundles	Radial; 2-6 (tetrarch)	Radial; polyarch (6+)	Ring; open; collateral	Scattered; closed; collateral
Cambium	Present (secondary growth)	Absent	Present (open)	Absent (closed)
Pith	Small / absent	Large and distinct	Distinct, large	Not distinct
Xylem position	Exarch	Exarch	Endarch	Endarch
Secondary growth	Present	Absent	Present	Absent

Dorsiventral (Dicot) Leaf: Upper epidermis (no stomata, thick cuticle) → Palisade mesophyll (columnar, dense chloroplasts, main photosynthesis) → Spongy mesophyll (loose, gas exchange) → Lower epidermis (stomata abundant, kidney-shaped guard cells).

Isobilateral (Monocot) Leaf: Both epidermis have stomata (amphistomatic); mesophyll NOT differentiated; Kranz anatomy (bundle sheath) for C4 photosynthesis; bulliform cells help in leaf rolling during drought.

UNIT IV | General Idea of Plant Physiology (12 Lectures)

1.1 Water Relations — Diffusion, Osmosis, Water Potential

Water Potential (Ψ) = Ψ_s + Ψ_p | Pure water $\Psi = 0$ | Solutes make Ψ negative

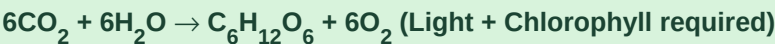
Term	Definition	Key Point
Diffusion	Movement of molecules from high to low concentration	Passive; no membrane needed; reaches equilibrium
Osmosis	Diffusion of WATER through a semipermeable membrane	From dilute (high Ψ) to concentrated (low Ψ) solution
Imbibition	Absorption of water by solid colloids (no solution formed)	Generates enormous imbibitional pressure — seed germination
Turgidity	Cell fully hydrated; turgor pressure maximum	Maintains shape of soft plant parts
Plasmolysis	Cell in hypertonic solution; membrane shrinks away from wall	Exosmosis — can be reversed (deplasmolysis)

1.2 Transpiration and Ascent of Sap

Type	Location	Percentage
Stomatal	Through stomata (guard cells)	80 – 90 %
Cuticular	Through cuticle on leaf surface	10 – 20 %
Lenticular	Through lenticels (bark pores)	< 1 %

Cohesion-Tension Theory (Dixon & Joly, 1894 — most accepted): Transpiration from leaves creates **tension** (negative pressure) in xylem. **Cohesion** (H-bonds between water molecules) and **adhesion** (water to xylem walls) maintain a continuous water column pulled from roots to leaves — can reach heights >100 m in tall trees.

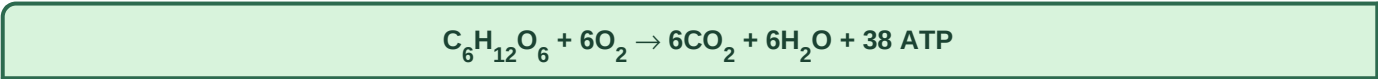
1.4 Photosynthesis



Stage	Location	Key Events	Products
Light Reactions (Photo reactions)	Thylakoid membranes (grana)	Light absorbed; water photolysis; PS I & PS II involved	O ₂ released; ATP + NADPH formed
Calvin Cycle (Dark/C ₃ reactions)	Stroma	CO ₂ fixed by RuBP (Rubisco enzyme); PGA formed	G3P → Glucose; RuBP regenerated

C4 Pathway (Hatch-Slack)	Mesophyll + Bundle sheath	OAA (4C) first product; Kranz anatomy; high temp efficiency	Less photorespiration; more sugar
CAM Pathway	Desert succulents	Stomata open at NIGHT only; CO2 stored as malic acid	Conserves water in arid conditions

1.5 Respiration



Stage	Location	Inputs → Outputs	ATP yield
Glycolysis (EMP)	Cytoplasm	Glucose → 2 Pyruvate + 2 NADH	2 ATP (net)
Pyruvate Oxidation	Mitochondrial matrix	Pyruvate → Acetyl-CoA + CO2 + NADH	0 ATP
Krebs Cycle (TCA)	Mitochondrial matrix	Acetyl-CoA → CO2 + NADH + FADH2	2 ATP
ETC + Ox. Phosphorylation	Inner mito. membrane (cristae)	NADH/FADH2 → H2O; proton gradient → ATP synthase	32–34 ATP

RQ (Respiratory Quotient) = CO₂ released / O₂ absorbed: Carbohydrates = 1.0 | Fats < 1 (0.7) |
Proteins ≈ 0.9 | Organic acids > 1



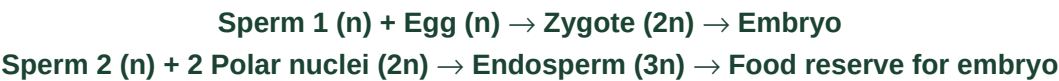
UNIT V | Plant Reproduction, Growth and Propagation (12 Lectures)

1.1 Plant Propagation — Types of Propagules

Propagule	Type	Example
Rhizome	Underground stem	Ginger, Turmeric, Lotus
Tuber	Underground stem (swollen tip)	Potato
Bulb	Modified stem with scale leaves	Onion, Garlic
Corm	Solid underground stem	Colocasia, Saffron
Runner	Sub-aerial stem on soil	Grass, Strawberry, Oxalis
Sucker	Underground stem (oblique)	Banana, Mint, Chrysanthemum
Bulbil	Fleshy bud	Agave, Garlic
Foliar buds	Leaf margin buds	Bryophyllum
Cutting (artificial)	Stem/leaf/root piece	Rose, Hibiscus, Begonia
Grafting (artificial)	Stock + Scion union	Mango, Apple, Rose

1.2 Sexual and Asexual Reproduction

Double Fertilization (unique to Angiosperms — discovered by Nawaschin, 1898):



Pollination Type	Agent	Adaptations / Examples
Autogamy	Self (same flower)	Cleistogamy; pollen shed before flower opens
Anemophily	Wind	Light pollen, feathery stigma — Wheat, Maize, Grass
Entomophily	Insects	Colorful petals, nectar, scent — Rose, Sunflower, Hibiscus
Hydrophily	Water	Reduced perianth, non-sticky pollen — Vallisneria, Hydrilla
Ornithophily	Birds	Red/orange flowers, copious nectar — Bombax, Bottlebrush

1.3 Plant Hormones and Role of Elicitors

Hormone	Site Produced	Primary Effects
Auxin (IAA)	Shoot apex, young leaves	Cell elongation, root initiation, apical dominance, fruit development
Gibberellin (GA)	Young leaves, seeds	Stem elongation, seed germination, bolting, fruit growth

Cytokinin (CK)	Root apex, endosperm	Cell division, shoot differentiation, delays leaf senescence
ABA (Absciscic acid)	Old leaves, fruits	"Stress hormone" — stomata closure, dormancy, abscission
Ethylene (C2H4)	Ripening fruit, aging tissue	Fruit ripening, senescence, abscission of leaves/flowers

Elicitors are signal molecules (biotic — from pathogens; abiotic — UV, heat, wounding) that trigger plant defence: secondary metabolite production (phenols, terpenoids, alkaloids) and Systemic Acquired Resistance (SAR). Used in agriculture to reduce pesticide use.

1.4 & 1.5 Gametophyte, Sporophyte and Alternation of Generations

Plant Group	Dominant Phase	Gametophyte	Sporophyte
Algae	Gametophyte (n)	Free-living, large	Reduced or absent
Bryophytes (Mosses)	Gametophyte (n)	Free-living leafy plant	Dependent on gametophyte
Pteridophytes (Ferns)	Sporophyte (2n)	Small prothallus	The fern plant we see
Gymnosperms	Sporophyte (2n)	Reduced (pollen/embryo sac)	The tree / shrub
Angiosperms	Sporophyte (2n)	Pollen (3-celled) / Embryo sac (7-celled)	The flowering plant

Cycle: Sporophyte (2n) →[Meiosis]→ Spores (n) →[Mitosis]→ Gametophyte (n) →[Gametes + Fertilization]→ Zygote (2n) → New Sporophyte

QUICK REFERENCE | Summary & Exam Tips

Unit	Topic	Must-Know Points
I	Bhartiya Gyan Parampara	Vedas plant classification; Parashar, Charak, Sushrut contributions; Folk medicine plants
II	Morphology	All root/stem/leaf modifications with scientific names; Fruit classification; Pollination types
III	Anatomy	Cell organelles (plant-specific); Tissue types; Dicot vs Monocot root/stem/leaf differences
IV	Physiology	Osmosis & water potential equations; Cohesion-tension theory; Photosynthesis stages; Respiration ATP yield
V	Reproduction	Double fertilization; Plant hormones; Elicitors; Alternation of generations cycle

Exam Tips:

- Draw **labelled diagrams** for: plant cell, CS of root/stem/leaf (dicot & monocot), flower.
- Learn all **modifications with at least 2 examples each** and their scientific names.
- Know the **differences between dicot and monocot** structures — common exam question.
- Memorise **photosynthesis and respiration equations** along with stages and locations.
- Prepare a table of **Indian Rishis and their contributions** — appears in Part A questions.
- Understand **double fertilization** — unique to angiosperms and often asked for 5 marks.
- For elicitors, link to **practical use in agriculture** and SAR concept.
- Revise **alternation of generations** with a clear cycle diagram.

Best of Luck! ■ Study Smart — Nature is the Best Teacher